Baseline Dense Retrieval versus Cross-Encoder Re-Ranking on the FIA 2026 Formula 1 Sporting Regulations

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Code and results: https://github.com/abhinavborah/rag-rerank-eval

Abstract. **Two pipelines are compared over the same 375 chunks of one regulatory PDF, with the same prompt and language model. Pipeline 1 passes the top 3 of 10 chunks retrieved by a bi-encoder. Pipeline 2 re-orders the same 10 with a cross-encoder and passes the new top 3. On 30 queries fixed before any run, re-ranking brought a gold chunk into the top 3 for no query and removed one for two. Restored cases were found by later searches, which are labelled as such: of 19 model-written questions with the gold chunk at baseline ranks 4 to 10, 15 were restored by rank and 11 changed from a wrong or "not covered" answer to a correct one, while re-ranking all 150 gained 15 and lost 7. Re-ranking 10 pairs costs about 69 ms against 7 ms for retrieval.**

I. Setup

The document is the FIA 2026 Formula 1 Regulations, Section B (Sporting), Issue 09, 1 October 2026 [1] (SHA-256 `5e156f50...4b85`). Pages 4 to 94 are indexed; the cover, colour legend and contents (pages 1 to 3) and the approved changes for 2027 to 2029 (pages 95 to 101) are left out. Struck-out text (a magenta line in the PDF), page headers and footers and appendix row numbers are removed, and annotation text is kept. Each numbered clause is one chunk with its article, section and clause number in front, split at 400 tokens, giving 375 chunks.

The bi-encoder is `BAAI/bge-small-en-v1.5` [2], [3] with exact inner-product search in FAISS [4]. The cross-encoder is `cross-encoder/ms-marco-MiniLM-L-6-v2` [5], [6], [7]. The generator is `claude-haiku-5-5`, accessed on 18 and 19 October 2026, with one system prompt for both pipelines: answer only from the numbered passages, cite them, and say so if the document does not cover the question. Its output is not deterministic, so every answer is saved.

The *main set* has 30 queries with gold chunks written before any run (3 control, 6 keyword trap, 9 exception, 7 negation, 4 scattered, 1 out of scope). Gold is the smallest set of chunks that gives a complete answer. Metrics are hit@3 (any gold chunk in the top 3), recall@3 (gold chunks in the top 3 divided by the smaller of 3 and the number of gold chunks) and MRR (reciprocal rank of the best-placed gold chunk, 0 if none is among the 10). Q09 is scored on retrieval only because its tyre table is garbled in text extraction. A query is *restored* when its gold chunk is outside the baseline top 3 and inside the re-ranked top 3. All other query sets below were designed after the main results were known.

TABLE I
Main set, baseline / re-ranked (29 scored queries).

Group	n	hit@3	recall@3	MRR
control	3	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
keyword trap	6	0.67 / 0.50	0.67 / 0.50	0.47 / 0.46
exception	9	0.89 / 0.89	0.70 / 0.67	0.70 / 0.83
negation	7	0.86 / 0.86	0.79 / 0.79	0.62 / 0.62
scattered	4	1.00 / 1.00	0.75 / 0.75	0.75 / 1.00
all	29	0.86 / 0.83	0.75 / 0.71	0.67 / 0.74

II. Architecture comparison

A bi-encoder embeds the query and each chunk independently and scores them by cosine similarity [8]. Chunk vectors are computed once and indexed, so a query costs one encoding and one search. The price is that a chunk is compressed into one vector before the query is known, and words such as "not" or "except" shift that vector only slightly. A cross-encoder reads the query and chunk as one sequence, so self-attention relates every query token to every chunk token [9]. It can see that a negation applies to what the query asks about. The score depends on the pair, so nothing can be precomputed.

III. Main set and failure modes

Re-ranking moved the best gold chunk up for 8 queries and down for 3 (a sign test gives $p \approx 0.23$, so only counts are reported). It brought no additional gold chunk into the top 3 and removed a gold chunk from it for two queries (Q05 and Q14; Q14 kept another gold chunk there). For Q04, Q12 and Q23 no gold chunk was among the 10 retrieved. The main set therefore has no restored query.

TABLE II
FAILURE CATEGORIES ("OUT" = NOT RETRIEVED).

Category	Cases
Keyword trap	Q04 ("mechanics" for "team personnel"): `B5.5.4` out → out. Q07: `B5.14.2#2` 8 → 7; Answer 2 reports a tyre penalty from another clause that also says "suspended and cannot be re-started".
Exception	Q12: both gold out → out. Q15: exception clause `B3.5.4` 10 → 7, still outside the top 3.
Negation	Q21: gold at rank 1, both answers hedge because the TTCS definition is not retrieved. Q23: `B3.3.3` out → out.
Buried (4 to 10)	Q25 (typed as negation, best gold chunk at rank 2): the second gold chunk `B1.9.5#1` stays at 4 and both answers hedge. Restored cases: next section.

Only Q07 had its best gold chunk at baseline ranks 4 to 10, the only position from which a 10-candidate re-ranker can promote it into the top 3.

Table II lists the cases for each failure category in the assignment. Ranks are baseline → re-ranked.

IV. QUERIES WHERE RE-RANKING RESTORES RELEVANCE

The assignment asks for at least two such queries, and the main set has none, so further queries were searched for under rules written and committed before each run, with every candidate logged. They are reported separately from the main set and are not merged into its metrics.

Hand-written batches. A batch of 14 queries on near-duplicate clauses restored one, X10 ("What can the stewards do about an incident during Qualifying?"): `B1.9.4` moved from rank 6 to 1. Answer 1 says the passages do not address Qualifying. Answer 2 gives the rule (lap times deleted or grid places dropped, penalties served in the Race, Sprint Qualifying penalties applied to the Sprint grid) but expands LTCS wrongly as "Lap Time Competition Session". A second screen of 18 hand-written candidates kept one with gold at baseline ranks 4 to 10 (rank 10 to 5, outside the top 3), so it was not restored.

Model-written questions. To remove the author's choice of queries, 150 chunks were sampled at random (seed 20261019) and the generator wrote one question for each, with the source chunk as gold. The instruction asked for everyday wording instead of the regulation's defined terms, which deliberately targets vocabulary mismatch. Under a cosine-only screen the gold chunk was at rank 1 for 94 questions, ranks 2 to 3 for 25, ranks 4 to 10 for 19 and outside the top 10 for 12. The 19 were run through both pipelines once. The gold chunk entered the top 3 for 15, moved up for 18 and down for 1. Read against the clause text, in 11 of the 15 Answer 1 says "not covered" or is wrong and Answer 2 is correct (Table III). In the other four the rank changed but the answer did not, or the gold label is weak.

By the assignment's categories, six of the eleven involve terminology that differs between query and clause, three

TABLE III
MODEL-WRITTEN QUESTIONS WHERE THE ANSWER CHANGED FROM WRONG OR "NOT COVERED" TO CORRECT.

Id	Rank	Answer 1	Answer 2
Z027	6 → 1	halted start not described	cars enter Pit Lane behind the Safety Car in order
Z042	7 → 1	not covered	60 operational personnel, unlimited near the start
Z050	10 → 2	not covered	who may be on the track
Z059	4 → 2	Q1, Q2 not mentioned	Q2 losers 11 to 16, Q1 losers 17 to 22
Z064	5 → 1	not covered	no work until the penalty period ends
Z074	5 → 1	not covered	parc fermé from first Pit Lane exit
Z115	8 → 3	wrong overtaking rule	no overtaking until the Control Line
Z128	4 → 3	not covered	25 minutes, receipt acknowledged
Z130	10 → 1	hours not found	one car, 09:00 to 19:00
Z131	5 → 1	not covered	returns after "LAPPED CARS MAY NOW OVERTAKE"
Z135	7 → 3	position not specified	back of the Fast Lane line, in order of arrival

are exceptions or corner cases (Z050, Z064, Z135) and two are look-alike distractors (Z115, Z130). None is a negation query. These categories were assigned after the run. Caveats: the 19 were selected for gold at ranks 4 to 10, so a move up is partly expected, and if the re-ranker ordered the 10 chunks at random, 15 or more of 19 would have probability about 1.5×10^{-5}, which rules out chance but says nothing about losses. Re-ranking all 150 questions without answers gained 15 and lost 7 (2 of the 94 with gold at rank 1 and 5 of the 25 at ranks 2 to 3), so hit@3 rose from 0.79 to 0.85, while on the 131 questions that were not selected it fell from 119 to 112. The questions are model written. Across the main set and the three later query sets, 22 of 211 queries had the best gold chunk at ranks 4 to 10, and 16 of those were restored by rank and 12 by answer.

Outside the assignment design, re-ranking 50 candidates instead of 10 (retrieval and re-ranking only) lifted Q23 from cosine rank 40 to 1 and Q12 from 19 to 2, which are recall failures at 10. It pushed Q04 (11 to 21), Q05 (3 to 25) and a follow-up query (15 to 35) down, so hit@3 on the main set rose from 25 to 26 of 29 (0.86 to 0.90), a net gain of one query.

V. ANSWER 1 AGAINST ANSWER 2

Among the answers read against the clause text, one invented detail was found (the LTCS expansion in X10 Answer 2). The usual failure of Answer 1 is omission: "not covered" when the critical chunk is outside the top 3 (X10, Z042, Z050, Z064, Z074, Z128, Z131, Z135). A rule applied to the wrong situation is the other failure (Z115 in Answer 1). In the main set one case is a clear

TABLE IV
MEDIAN LATENCY PER QUERY.

Stage	Time
Pipeline 1: embed query, search	6.8 ms
Pipeline 2: re-rank 10 pairs	68.9 ms
Pipeline 2 total	75.7 ms
Re-rank 50 pairs (extra)	347 ms
Answer generation (median of 60)	3.25 s

degradation by re-ranking: for Q05 the gold chunk fell from rank 3 to 9, and Answer 2 says the passages do not cover jump starts while Answer 1 lists the penalties. For Q14 Answer 2 gives the right verdict but cites an energy-mode clause instead of the overtaking rule, and Answer 1 hedges. For Q07 Answer 2 reports a tyre penalty from a clause that contains the query's words, and Answer 1 does not answer. The first Answer 2 for Q26 was cut off at the output budget, and a rerun at 2000 tokens listed the penalties, so that difference is confounded with the budget and is not counted as a degradation. Five saved main-set answers and six model-written answers reached the 800-token output budget because hidden reasoning counts toward it. Both answers of each affected pair were regenerated at 2000 tokens as a labelled supplement, the originals remain the record, and the judgements for Z027 and Z059, which rest on cut-off answers, hold under the rerun.

VI. Latency

Table IV gives warm per-stage medians over the 30 main queries, 5 runs each, on an Apple M4 Pro CPU with torch 2.14.1.

Re-ranking takes about 10 times as long as retrieval, at about 6.9 ms per pair. Generation is much slower than both. The first three rows are medians over 30 queries by 5 runs, and the total is the sum of the stage medians. The 50-pair row is a median over 43 queries by 3 runs, and the generation row is the median of the logged times of the 60 original calls.

VII. Why not cross-encode the whole collection

The cost grows linearly with the number of chunks, since each needs a forward pass with the query. At 6.9 ms per pair the 375 chunks here would need about 2.6 s per query, against 7 ms for the bi-encoder, and one million chunks would need about 1.9 hours. The bi-encoder computes chunk vectors once, so each chunk costs one dot product, and approximate indexes make the search sublinear for large collections (the exact index used here scans every vector). The cross-encoder cannot precompute anything because its input is the pair. The two-stage design therefore uses the cheap model for recall over everything and the expensive model for precision over a short list. The 50-candidate result shows the trade-off: more candidates rescued two queries and lost one, at five times the re-ranking time.

VIII. Limitations

- The query sets are small and cover one document, so differences of one or two queries are not evidence of a gain or a loss.
- All later sets were designed after the main results were known, and the model-written questions share wording with the chunks and were written by the model that answers them.
- The cross-encoder was trained on web passages and may transfer less well to regulations, although zero-shot benchmarks report strong averages for such re-rankers [10]. Several hedged answers trace instead to the TTCS definition never being retrieved, a limit of clause-level chunks.
- The generator is not deterministic. When both pipelines pass the same chunks in a different order, an answer difference is partly noise.

IX. Supplementary material

Rank-shift tables, the 50-candidate table and verbatim answer excerpts are in the repository, with all links fixed to commit `598a0d1`: results/rank_shift_summary.md, the notebook with outputs, results/query_log.jsonl, results/gen_run_log.jsonl and README.md (source changes, models, how to run).

References

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